

Technical Publication

Diesel Engine
8V 183 TD 13
Base: OM 442 LA
- EURO 2 -

Operating Instructions
M015198/10E

.

DETROIT DIESEL



Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

Diese Veröffentlichung einschließlich aller ihrer Teile ist urheberrechtlich geschützt. Jede Verwertung oder Nutzung bedarf der vorherigen schriftlichen Zustimmung der MTU Friedrichshafen GmbH. Das gilt insbesondere für Vervielfältigung, Verbreitung, Bearbeitung, Übersetzung, Mikroverfilmungen und die Einspeicherung und / oder Verarbeitung in elektronischen Systemen, einschließlich Datenbanken und Online-Diensten. Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen. Änderungen bleiben vorbehalten.

Printed in Germany

© 2006 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Imprimé en Allemagne

© 2006 Copyright MTU Friedrichshafen GmbH

Tout droit réservé pour cet ouvrage dans son intégralité. Toute utilisation ou exploitation requiert au préalable l'accord écrit de MTU Friedrichshafen GmbH. Ceci s'applique notamment à la reproduction, la diffusion, la modification, la traduction, l'archivage sur microfiches, la mémorisation et / ou le traitement sur des systèmes électroniques, y compris les bases de données et les services en ligne.

Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

© 2006 Copyright MTU Friedrichshafen GmbH

Esta publicación se encuentra protegida, en toda su extensión, por los derechos de autor. Cualquier utilización de la misma, así como su reproducción, difusión, transformación, traducción, microfilmación, grabación y/o procesamiento en sistemas electrónicos, entre los que se incluyen bancos de datos y servicios en línea, precisa de la autorización previa de MTU Friedrichshafen GmbH.

El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

Nos reservamos el derecho de introducir modificaciones.

Stampato in Germania

© 2006 Copyright MTU Friedrichshafen GmbH

Questa pubblicazione è protetta dal diritto d'autore in tutte le sue parti. Ciascun impiego o utilizzo, con particolare riguardo alla riproduzione, alla diffusione, alla modifica, alla traduzione, all'archiviazione in microfilm e alla memorizzazione o all'elaborazione in sistemi elettronici, comprese banche dati e servizi on line, deve essere espressamente autorizzato per iscritto dalla MTU Friedrichshafen GmbH.

Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

Impresso na Alemanha

© 2006 Copyright MTU Friedrichshafen GmbH

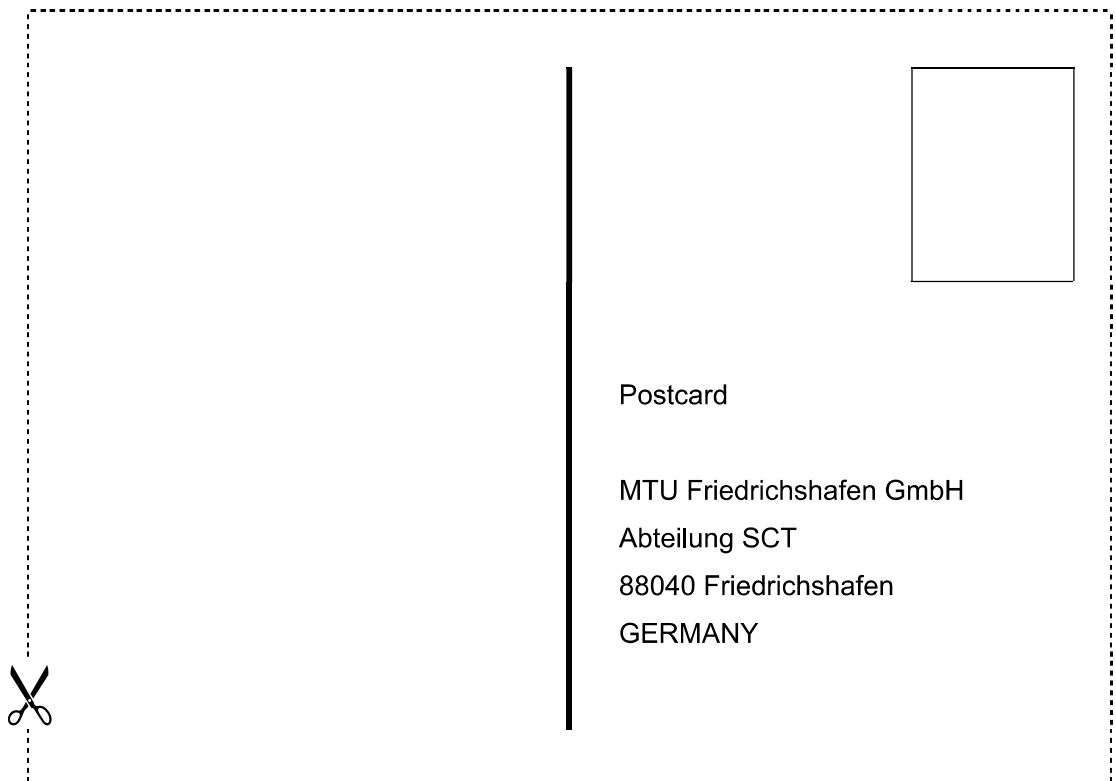
A presente publicação, inclusive todas as suas partes, está protegida pelo direito autoral. Qualquer aproveitamento ou uso exige a autorização prévia e por escrito da MTU Friedrichshafen GmbH. Isto diz respeito em particular à reprodução, divulgação, tratamento, tradução, microfilmagem, e a memorização e/ou processamento em sistemas eletrônicos, inclusive bancos de dados e serviços on-line.

Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

Amendment Service!

Any changes in contents will be sent to you in the form of an amendment, provided you complete the reverse of this receipt card and return it to us.



Please use block capitals!



Receipt

Tick one box only!

- ☐ Update as (PDF) file by E-mail
- ☐ Update as (PDF) file as CD-ROM
- ☐ Update in hardcopy form

Publication number: **M015198/10E**

Name

Manufacturer

Department E-Mail

Fax Telephone

Street

Postal box number

(Postal code) City

Country

Preliminary Remarks/Important Information

Operation and maintenance are important factors in determining an engine's availability for operation and length of service life. These Operating Instructions are intended to provide clear answers to important questions regarding operation and maintenance.

We strongly recommend that the specified maintenance tasks be carried out carefully and on schedule. In order not to jeopardize warranty claims, conditions for engine operation which deviate from the normal operating conditions must be taken into consideration.

In event of questions concerning customer service, repairs or spare parts for the engine, please first contact the supplier of the equipment or component. Always state the complete engine No. given on the engine nameplate. If necessary, the supplier will use the MTU or MERCEDES-BENZ Service.

Furthermore, MTU's Product Support Service is available in an advisory capacity.

**MTU Motoren- und Turbinen-Union
Friedrichshafen GmbH**
88040 Friedrichshafen/Germany
Telephone (07541) 90-0
Telex 7 34 280-0 mt d
Telefax (07541) 90 - 39 28

Notes for the user

This Manual consists of 10 Sections.

The Sections are listed on this page.

Each Section has its own detailed Table of Contents.

Using the table of Contents to trace an entry

- ↓ Table of Contents (general)
- ↓ Contents
- ↓ Summary/Page

Using the Alphabetical Index to trace an entry

- ↓ Alphabetical keyword
- ↓ Heading/Page

Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

1 Safety

Safety and accident prevention measures and instructions

2 Description

System and component descriptions

3 Operation

Initial operation, starting, monitoring and shutdown

4 Maintenance

Maintenance intervals, schedules and frequency charts, maintenance tasks

5 Cleaning, Preservation

Cleaning and preservations specifications

6 Practical Advice

Faults, causes and remedies, test and assembly work, injector replacement

7 Fluids and Lubricants

Safety instructions, diesel fuel, engine oils and coolants

8 Technical Data

Engine data, test values and settings, tightening torques

9 Keyword Index

Alphabetical index of key terms

10 Appendix

Contents	Page
1.1 General	5
1.2 Correct use.....	5
1.3 Symbols.....	5
1.4 Personnel requirements	6
1.5 Conversion and alteration of the engine	6
1.6 Transportation	6
1.7 Installation.....	7
1.8 Organizational measures	7
1.9 Safety measures on engines with electronic engine management ECS - 183/2	7
1.10 MTU and Mercedes-Benz original components....	8
1.11 Safety and emergency propulsion program	9
1.12 Automatic emergency shutdown	9
1.12.1 EHAB (Electro-hydraulic shutdown unit).....	9

1.1 General

The engine is a state-of-the-art product and conforms with all applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of

- incorrect use
- operation by unqualified personnel or personnel not trained by Mercedes-Benz or MTU
- incorrect alteration or conversion
- safety instructions being disregarded

1.2 Correct Use

The engine is intended solely for the use specified in the contract. Any other use is considered improper use. MTU and Mercedes-Benz will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Liability for any damage resulting from this use is borne solely by the manufacturer of the finished machine/vehicle.

These Operating Instructions must be followed. The maintenance and care specifications must be followed.

1.3 Symbols

	Risk of injury!
	This symbol is used with all safety instructions concerning a hazard which, if disregarded, may result in immediate danger for human beings.

NOTE:

Safety instructions are given primarily in cases where, if disregarded, there is the risk of material, functional and environmental damage.

Instructions for environmental protection are also marked.

Other instructions provide useful information on engine operation and maintenance.

1.4 Personnel Requirements

Work on the engine must be carried out only by appropriately qualified personnel, or personnel trained by MTU or Mercedes-Benz, including personnel of workshops authorized by MTU or Mercedes-Benz.

Responsibilities of the operating, maintenance and repair personnel must be specified.

The specified legal minimum age must be respected.

1.5 Conversion and Alteration of the Engine

Unauthorized alterations to the engine may affect its operation and safety. No liability for resultant damage is accepted.

Similarly, manipulation of the injection and governor systems may affect the performance and exhaust emissions of the engine. Factory settings and fulfillment of statutory environmental conditions are then no longer guaranteed.

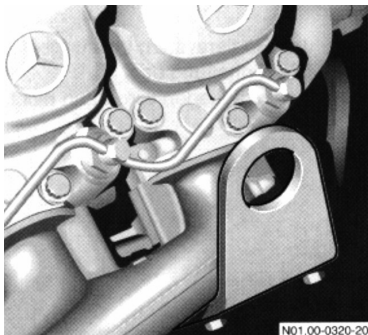
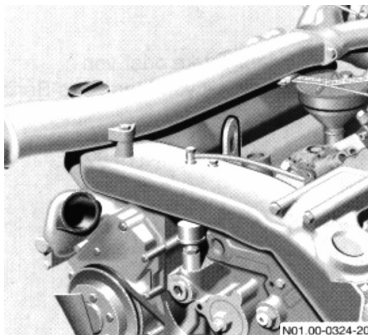
1.6 Transportation

Use only the transport equipment approved by MTU or Mercedes-Benz. Follow lifting specifications.

NOTE:

The engine must only be transported in installation position.

Use only the engine's own lifting fixtures.



1.7 Installation

The engine is intended solely for the use specified in the contract. See Sections "Proper Use" (Page 5) and "Conversion and Alteration of Engine" (Page 6).

These Operating Instructions contain data for installation in the "Technical Data" Section. In event of queries, please contact your MTU or Mercedes-Benz Service.

1.8 Organizational Measures

These operating instructions must be issued to the personnel involved with operating or working on the engine and kept at the engine's site of operation, where they should be at all times accessible.

The personnel must be instructed on engine operation by means of the Operation Manual, and in particular the safety instructions must be explained. This is especially important for personnel who work on the engine only on an occasional basis.

In addition to the operating instructions, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed.

1.9 Safety Measures for Engines with Electronic Engine Management ECS - 183/2

All adjustable parameters (limits and trip points for engine control and monitoring) must be coordinated with the responsible department VTO at MTU Friedrichshafen.

It is essential to carry out the following safety measures to prevent damage to the engine, components and wiring, as well as injury to personnel.

NOTE:

- Start engine only when batteries are securely connected.
- Do not disconnect batteries when engine is running.
- Start engine only when speed transmitters are installed and connected.
- Do not use boost chargers to start engine. Start assistance only with separate batteries.
- To boost charge the batteries, the battery terminals must be removed. See operating instructions of boost chargers.
- For electro-welding work, the batteries must be disconnected and the two cables (+ and -) securely connected together. Separate plug-in connections of control units.
- The plug-in connections of the control units must only be disconnected and connected with the electrical system switched off.
- Incorrect polarity of the control unit supply voltage (e.g. incorrect polarity of the batteries) can result in destruction of the control units.

- It is essential to tighten the union nuts of the electric connectors (controlling mechanism for injection pump sliding-sleeve, temperature and pressure sensors) to specified torque - see "Technical Data".
- If the two speed transmitters are interchanged in the engine control housing, this may result in damage from overspeeding during the starting sequence.
- If temperatures exceeding 80 °C are expected (e.g. drying stove), the control units must be removed.

1.10 MTU and Mercedes-Benz Original Components

The original components are subjected to very stringent quality checks and guarantee maximum function, safety and value retention. Each component is specially designed, produced, selected and approved for MTU and Mercedes-Benz.

We must therefore reject any responsibility for damage resulting from the use of components and accessories which do not satisfy these conditions.

NOTE:

For economical repair in the course of recycling, MTU and Mercedes-Benz exchange units and components are offered, which come with the same quality standards and guarantee as new components.

In the Federal Republic of Germany and some other countries, certain components, for example safety components, are permitted for installation and conversion only if the applicable legal requirements are satisfied. This requirement is fulfilled in every case by MTU and Mercedes-Benz original components.

If other components which have not been tested and approved by MTU or Mercedes-Benz are used - even if, in individual cases, they have been accepted or officially authorized - this cannot be assessed or guaranteed, despite continuous market observation.

Warranty claims may therefore be restricted.

When ordering original components, and in event of technical queries, always state the engine number with type test designation.

NOTE:

In order to determine exactly the design status, the complete engine number and engine data card (see Page 14) are always required.

1.11 Safety and Emergency Propulsion program

The engines have an electronics system which monitors both the engine and itself (self-diagnosis).

As soon as the system detects a fault, one of the following measures is automatically initiated, according to evaluation of the fault:

- Display of faults during operation by means of a fault code display installed in connection interface 1

Detected faults, even temporary ones, are stored by the system in decimal code and issued in binary code.

- Changeover to suitable substitute function for further, though unrestricted engine operation (e.g. reduced speed).
- Immediate engine shutdown, if necessary for safety reasons. According to fault evaluation, the engine is shut down by reducing delivery to zero, and emergency shutdown is initiated.

NOTE:

The fault must be rectified immediately by the responsible MTU or Mercedes-Benz Service.

1.12 Automatic Emergency Shutdown

Any required emergency engine shutdown is effected by the electronic engine management system ECS-183/2 via an electro-hydraulic shutdown unit (EHAB).

1.12.1 EHAB (electro-hydraulic shutdown unit)

When activated, the EHAB reverses the delivery direction of the fuel delivery pump. By drawing off the fuel from the suction chamber of the injection pump, safe and rapid engine shutdown is ensured.

After every engine shutdown, the EHAB is checked for correct operation (run-on). For this reason, when the shutdown unit is actuated, the engine shutdown procedure is delayed by approx. 1.5 seconds.

Contents	Page
2.1 Type designation (Mercedes-Benz)	13
2.1.a Type designation (MTU)	13
2.2 Position of nameplate	13
2.3 Engine data card	14
2.4 Overview of OM 442 LA (View from free end, left engine side)	15
2.4.a Overview of OM 442 LA (View from free end, right engine side)	15
2.5 Description of engines	17
2.6 Electronic engine management ECS - 183/2	18
2.7 Arrangement of sensors	19

